



EXECUTIVE OVERVIEW

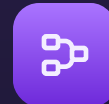
An AI assistant that reasons like a team and runs on your own machine.

Jarvis takes a request by voice or text and carries it all the way to a finished action on your behalf, whether that means replying to an email, booking a meeting, or pulling a market briefing. Instead of one model guessing an answer, a pipeline of specialised agents routes the request, plans the steps, checks its own work and scores the result before you ever see it. It runs a private model locally at no cost, and switches to a powerful cloud model with a single setting.



Understands & acts

Speak or type. Jarvis transcribes, reads intent, and performs the real action, not just a chat reply.



Reasons in stages

Five specialised agents plan, execute, critique and score every response before it reaches you.



Private by default

Runs a local model at \$0 with data kept on-device. One setting flips the whole stack to the cloud.



Connected to your world

Gmail, Calendar, markets, news, files, music and more. A dozen integrations the agents call on demand.

AT A GLANCE

● Local-first · \$0 to run

● 5-agent reasoning pipeline

● 12+ live integrations

● Voice + text, hands-free

● Local or cloud, one setting

Six layers, one continuous path

J.A.R.V.I.S

Multi-Agent AI Executive Assistant

Abdullah Khan Durrani
Final-year dissertation · designed & built solo

Local-first · runs at \$0

5-agent reasoning pipeline

Voice + text, hands-free

How a request travels: from your words to a real action taken on your behalf, handled by a team of specialised agents.

1

INTERFACE

You speak or type; a dashboard shows calendar, inbox and markets live.

Web Dashboard

text chat · live widgets

Voice I/O

hands-free · barge-in

2

VOICE PIPELINE

It listens, knows when you've finished talking, and replies in a natural voice.

Whisper STT

speech → text

Silero VAD

end-of-speech

ElevenLabs TTS

fast / hi-fi

Persona Layer

JARVIS voice

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ORCHESTRATOR · THE REASONING LOOP

A team of agents routes the request, plans the steps, checks its own work, scores the result, then writes the reply.

Router

intent · tier

Planner

builds task plan

Critic

re-plan check

Evaluator

scores output

Summariser

final reply

Memory

ChromaDB recall

re-plan if score low

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LLM BACKEND

One env var switches the whole stack between private local and cloud.

Ollama · Local

Llama 3.2 · private

Groq · Cloud

Llama 3.3 70B · Fly.io

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TOOLS & AGENTS

Where reasoning turns into action: a dozen integrations the agents call to actually get things done.

Gmail

read & reply in-thread

Calendar

events & scheduling

Web Search

live web answers

News

25+ sources

Markets

stocks & crypto

Sports

scores & tables

Weather

forecast & alerts

Spotify

play & control

Mac Control

apps · volume · files

File Manager

Desktop & Documents

Reminders

timed nudges

Prayer Times

daily times

6

FINEX · FINANCIAL STATEMENT SUB-AGENT

Reads company annual reports into a database, then answers questions over them with SQL + an LLM.

PDF Extraction

Postgres Store

SQL + LLM Analysis

grounded answers, with citations

Reasoning agents

Local model / voice

Cloud model

Tools & integrations

Financial sub-agent

Python · FastAPI · asyncio · ollama / Groq · ChromaDB · Whisper · Silero VAD · ElevenLabs · Postgres · Docker / Fly.io

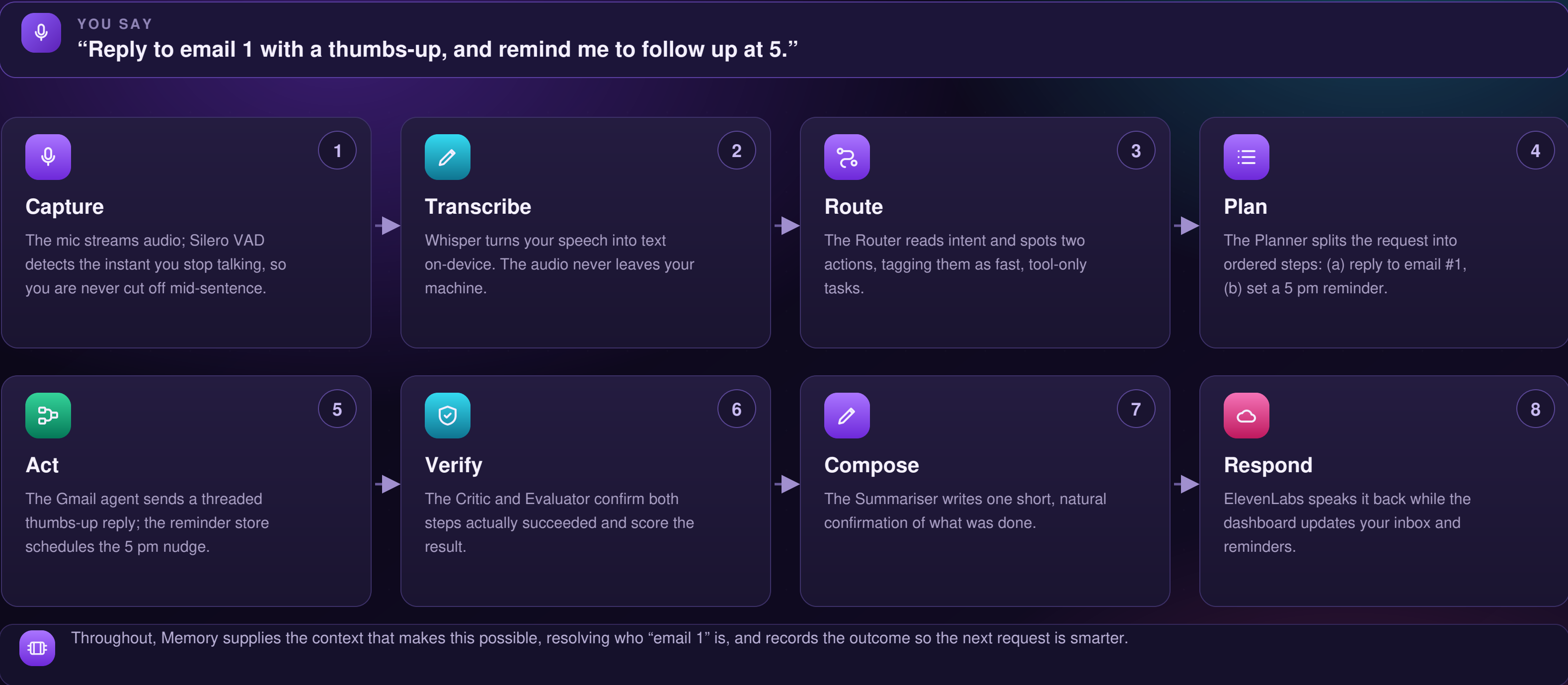
github.com/AbdullahKD

6 FinEx
Deep financial Q&A on demand.

Each layer hands its result to the next.
Memory is read when the request is routed
and written after every turn, so context
such as who you mean and what just
happened carries across the whole
conversation.

How a single request travels

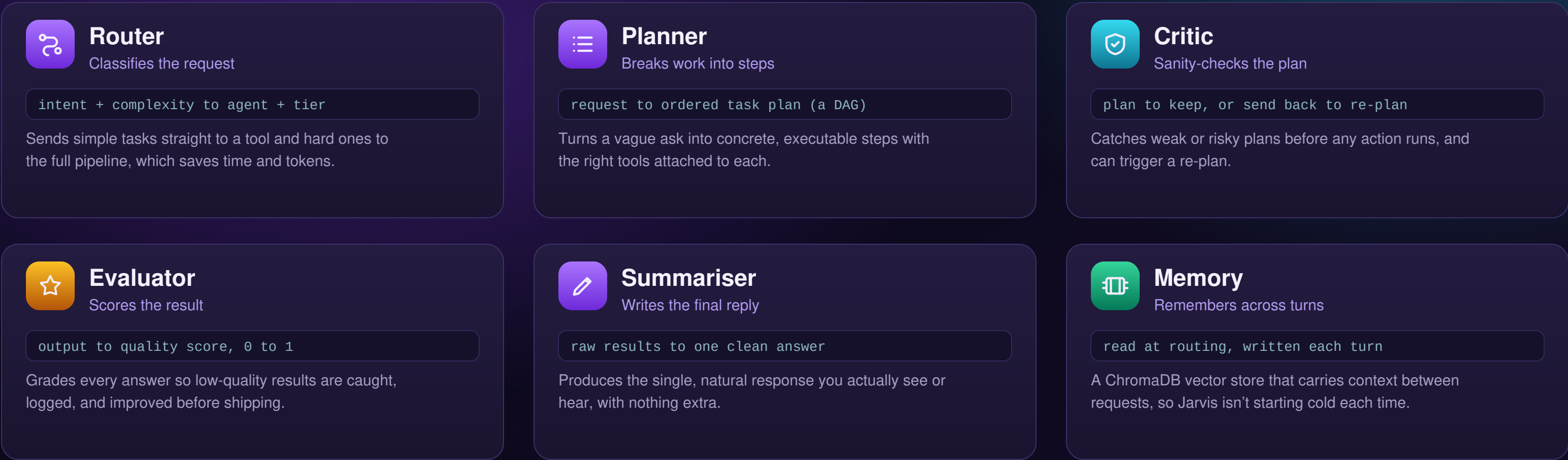
A worked example, traced through every layer, so the hand-offs and connections are explicit.



THE REASONING LOOP

A team of specialised agents

Rather than one model improvising, each agent owns one job. They run in sequence, and the chain can loop back to re-plan if quality is low.



THE ENGINE BEHIND THEM

Every agent calls one shared LLM. A single environment variable (LLM_BACKEND) swaps the engine for the entire stack, with no code changes.



CAPABILITIES

What it can actually do

Twelve live integrations the agents call to take action, plus a dedicated financial analyst, and how it deploys.

**Gmail**
Read, search and reply in-thread; sends real email for you.

**Calendar**
Create and read Google Calendar events; checks availability.

**Web Search**
Pulls live answers from the web for current questions.

**News**
Aggregates headlines from 25+ sources into briefings.

**Markets**
Live stock and crypto prices for NVDA, AAPL, BTC and more.

**Sports**
Scores and tables across football, basketball and cricket.

**Weather**
Current conditions and forecasts for any location.

**Spotify**
Play, pause, skip and control playback by voice.

**Mac Control**
Open apps, set volume and brightness, take screenshots.

**File Manager**
Find, read and organise files across Desktop, Docs, Downloads.

**Reminders**
Natural-language timed reminders, like “in 30 minutes”.

**Prayer Times**
Accurate daily prayer times for your location.

**FinEx, the financial statement sub-agent**
A specialist that ingests company annual reports (PDF) into a Postgres database, then answers financial questions over them by writing SQL and reasoning with an LLM, returning grounded answers with citations rather than guesses.

PDF Extraction

Postgres Store

SQL + LLM

Cited answer

**Deployment & privacy**
Runs locally on macOS for full privacy at zero cost, or deploys to the cloud (Fly.io + Groq + Neon Postgres) for a shareable URL. Mac-only features degrade gracefully in the cloud.

FastAPI · asyncio · ChromaDB · Whisper · ElevenLabs